

No disjoint-reiterative gap for mixing weighted shifts on c_0

Autonomous solution packet for arXiv:2412.19115

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Status: candidate partial result, likely valid. Every finite disjoint-hypercyclic tuple of mixing unilateral weighted backward shifts on $c_0(\mathbb{N})$ is disjoint reiteratively hypercyclic. Hence a weighted-shift counterexample to the source question cannot consist of mixing shifts. The unrestricted c_0 question remains open.

1 Source question and notation

Martin, Menet, and Puig asked in Question 5.1 of [1] whether two reiteratively hypercyclic operators can be disjoint hypercyclic but not disjoint reiteratively hypercyclic, and whether unilateral weighted shifts on $c_0(\mathbb{N})$ can provide such an example. Ri, Ju, and Kim quote the question as Question 1.1 and rule out the phenomenon on reflexive Banach spaces [2]. Since c_0 is nonreflexive, their result does not settle the weighted-shift part.

Let B_{w_s} be a unilateral weighted backward shift on $c_0(\mathbb{N})$, with bounded nonzero weights, and put

$$W_{j,n}^s = \prod_{\nu=1}^n w_{s,j+\nu} \quad (j, n \geq 1), \quad W_{j,0}^s = 1.$$

Thus $(B_{w_s}^n x)_j = W_{j,n}^s x_{j+n}$. Recall that B_{w_s} is mixing precisely when

$$|W_{j,n}^s| \longrightarrow \infty \quad (n \rightarrow \infty) \quad \text{for every fixed } j. \quad (\text{M})$$

For $A \subset \mathbb{N}$, $\overline{\text{Bd}}(A)$ denotes its upper Banach density.

2 Idea of the proof

Disjoint hypercyclicity allows one to prescribe, at one arbitrarily late time n , all weight-product ratios on any finite set of coordinates. Prescribe the data not only at $j \leq l$, but at the shifted coordinates $j + rM$. The cocycle identity then transports those prescriptions to every time $n + rM$. Repeating this with block lengths tending to infinity produces a common time set with upper Banach density at least $1/M$.

The remaining condition in the c_0 characterization is a cross-tail estimate. If two selected times differ by d , its ratio factors as a bounded same-time ratio divided by $W_{j,d}^s$. The selected set is M -separated, and mixing makes the denominator uniformly large for every $d \geq M$. This is the step unavailable under individual reiterative hypercyclicity alone.

3 Finite ratio interpolation

Lemma 1. *Suppose $B_{w_1}, \dots, B_{w_N}$ are disjoint hypercyclic on $c_0(\mathbb{N})$. For every $L, R \geq 1$, every nonzero scalars $c_{s,j}$ ($s \in [N]$, $j \in [L]$), and every $\eta > 0$, there is $n \geq R$ such that*

$$\left| \frac{W_{j,n}^s}{W_{j,n}^t} - \frac{c_{s,j}}{c_{t,j}} \right| < \eta \quad (s, t \in [N], j \in [L]). \quad (1)$$

Proof. Let x be a disjoint hypercyclic vector. Every tail of the diagonal orbit $((B_{w_1}^n x, \dots, B_{w_N}^n x))_{n \geq 0}$ is dense. Hence, for an arbitrarily small $\delta > 0$, some $n \geq R$ satisfies

$$|W_{j,n}^s x_{j+n} - c_{s,j}| < \delta \quad (s \in [N], j \in [L]).$$

All $c_{s,j}$ are nonzero. Taking δ sufficiently small and dividing the s and t coordinate inequalities gives (1), simultaneously for the finitely many ordered pairs. $\square$

4 The mixing c_0 theorem

Theorem 2. *Let $N \geq 2$. If $B_{w_1}, \dots, B_{w_N}$ are mixing unilateral weighted backward shifts on $c_0(\mathbb{N})$ and the tuple is disjoint hypercyclic, then the tuple is disjoint reiteratively hypercyclic.*

Proof. We verify Theorem 3.6 of [1] for the Furstenberg family of sets of positive upper Banach density.

Fix $l \geq 1$, nonzero target coefficients $a_{s,j}$ ($s \in [N]$, $j \in [l]$), and $\varepsilon > 0$. Replacing ε by $\min\{\varepsilon, 1/2\}$ causes no loss. Choose $K \geq 1$ so large that

$$\left| \frac{a_{s,j}}{a_{t,j}} \right| + 1 \leq K \quad (s, t \in [N], j \in [l]). \quad (2)$$

By (M), there is an integer $M > l$ such that

$$|W_{j,d}^s| > K/\varepsilon \quad (s \in [N], j \in [l], d \geq M). \quad (3)$$

For every $q \geq 1$, set $L_q = l + qM$. The sets of coordinates $\{j + rM : j \in [l]\}$, $0 \leq r \leq q$, are pairwise disjoint. Define nonzero scalars on $[L_q]$ by

$$b_{s,j+rM}^{(q)} = \frac{a_{s,j}}{W_{j,rM}^s} \quad (j \in [l], 0 \leq r \leq q), \quad (4)$$

and put $b_{s,i}^{(q)} = 1$ on unused coordinates. Lemma 1, with a tolerance small enough to absorb the finitely many factors $W_{j,rM}^s/W_{j,rM}^t$, gives a time n_q for which

$$\left| \frac{W_{j,n_q+rM}^s}{W_{j,n_q+rM}^t} - \frac{a_{s,j}}{a_{t,j}} \right| < \varepsilon \quad s, t \in [N], j \in [l], 0 \leq r \leq q. \quad (5)$$

Indeed, this follows from (4) and the exact cocycle identity

$$\frac{W_{j,n_q+rM}^s}{W_{j,n_q+rM}^t} = \frac{W_{j,rM}^s}{W_{j,rM}^t} \frac{W_{j+rM,n_q}^s}{W_{j+rM,n_q}^t}. \quad (6)$$

Choose the n_q successively so large that the blocks

$$A_q = \{n_q + rM : 0 \leq r \leq q\}$$

are mutually separated by at least M , and let $A = \bigcup_{q \geq 1} A_q$. Every two distinct points of A are at distance at least M . Moreover, for every window length H , every sufficiently long block A_q contains an interval of length H with at least $\lfloor (H-1)/M \rfloor + 1$ selected points. There are such blocks arbitrarily far out. Therefore

$$\overline{\text{Bd}}(A) \geq \lim_{H \rightarrow \infty} \frac{\lfloor (H-1)/M \rfloor + 1}{H} = \frac{1}{M} > 0. \quad (7)$$

We now check the three conditions in Theorem 3.6 of [1]. First, (M) and $n \rightarrow \infty$ along A give $|W_{j,n}^s| \rightarrow \infty$ for every s and $j \leq l$. Third, (5) is exactly the required same-time ratio approximation.

For the cross-tail condition, take $m, n \in A$, $s, t \in [N]$, and $j \leq l$ such that

$$j + m \in n + \{l + 1, l + 2, \dots\}.$$

Then $m > n$; because A is M -separated, $d = m - n \geq M$. From (2) and (5), after initially choosing the approximation tolerance below 1, all same-time ratios appearing below are bounded by K . The cocycle identity gives

$$\begin{aligned} \frac{|W_{j+d,n}^s|}{|W_{j,m}^t|} &= \frac{|W_{j,m}^s|}{|W_{j,m}^t| |W_{j,d}^s|} \\ &< \frac{K}{K/\varepsilon} = \varepsilon, \end{aligned} \quad (8)$$

where (3) was used in the last step. Thus all three conditions hold on the same A of positive upper Banach density. The cited characterization proves disjoint reiterative hypercyclicity. $\square$

5 Scope, upgrade audit, and verification

For two shifts Theorem 2 directly rules out the requested gap whenever both shifts are mixing. It does not settle the source question: individual reiterative hypercyclicity on c_0 need not provide (3). Attempts to replace mixing by chaos, frequent hypercyclicity, or reiterative hypercyclicity leave the same obstruction: the difference set of the common positive-upper-Banach-density set need not lie inside the times where every relevant product is large.

Eight focused upgrade routes were recorded in the accompanying attempt log. The final route tried to force entire arithmetic-block difference sets into the individual good-product sets; no available density principle justifies that step. The honest status is therefore a complete theorem for the mixing subclass, not a full answer.

The script `code/verify_block_identities.py` checks (6), the cross-tail factorization in (8), and the finite arithmetic-block densities for deterministic random complex weights. This is a sanity check, not part of the proof. The proof itself is symbolic and rests on the published c_0 characterization.

A bounded novelty search on August 11, 2026 used arXiv/web exact phrases for “mixing unilateral weighted shifts,” “disjoint reiteratively hypercyclic,” “ c_0 ,” and “upper Banach density.” It found [1], [2], and general background on disjoint hypercyclic weighted shifts, but no matching statement. Novelty confidence is moderate, not exhaustive.

References

- [1] O. Martin, Q. Menet, and Y. Puig, *Disjoint frequently hypercyclic pseudo-shifts*, J. Funct. Anal. 283 (2022), 109474; arXiv:2106.01409.
- [2] S.-U. Ri, H.-H. Ju, and J.-M. Kim, *A note on disjoint hypercyclicity for invertible bilateral pseudo-shifts on $\ell^p(\mathbb{Z})$* , arXiv:2412.19115v2.

Source evidence

Question 1.1 on page 2 of arXiv:2412.19115v2 reproduces Question 5.1 of Martin–Menet–Puig and records the reflexive-space partial answer.

powers of weighted shifts are pseudo-shifts, this means that as for the pseudo-shifts, there exist some tuples of invertible ones on $\ell^p(\mathbb{Z})$ with $1 \leq p < \infty$, such that they are disjoint hypercyclic.

Then, it is natural to ask what the conditions need for the tuple of bilateral pseudo-shifts on $\ell^p(\mathbb{Z})$ to be disjoint hypercyclic.

In this note we give a sufficient condition for bilateral pseudo-shifts on $\ell^p(\mathbb{Z})$ with $1 \leq p < \infty$ to be disjoint hypercyclic. Using this condition, we provide some examples of invertible pseudo-shifts which are disjoint hypercyclic, especially an example which one of them can not be described as powers of weighted shift, but they are disjoint hypercyclic.

At the end of this note we discuss the open problem from Martin, Menet and Puig [9]. In [9], they showed that two reiteratively hypercyclic unilateral pseudo-shifts on $\ell^p(\mathbb{N})$ with the same inducing map are disjoint hypercyclic if and only if they are disjoint reiteratively hypercyclic, and then they asked:

Question 1.1. (Question 5.1 of [9]) Is it possible to find two reiteratively hypercyclic operators T_1, T_2 which are disjoint hypercyclic but not disjoint reiteratively hypercyclic? Can T_1 and T_2 be taken as unilateral weighted shifts on $c_0(\mathbb{N})$?

We show that on a reflexive Banach space it is impossible, that is, reiteratively hypercyclic operators on a reflexive Banach space are disjoint hypercyclic if and only if they are disjoint reiteratively hypercyclic.

The organization of the paper is as follows. In Section 2, we give a sufficient condition for pseudo-shifts on $\ell^p(\mathbb{Z})$ with $1 \leq p < \infty$ to be disjoint hypercyclic and concrete examples. In Section 3, we discuss Question 1.1.

In the rest of Introduction, let us fix some notations and terminology.

Let X denote a separable infinite-dimensional Banach space over the real or complex scalar field $\mathbb{K}$ and $\mathcal{L}(X)$ denote the algebra of continuous linear operators on X . We say that an operator T in $\mathcal{L}(X)$ is *hypercyclic* if there exists a vector $x \in X$ (also called *hypercyclic*) such that its orbit $\text{Orb}(x, T) = \{T^n x : n \in \mathbb{N}\}$ is dense in X . Equivalently, there exists a vector $x \in X$ such that for any non-empty open set $U \subset X$ the return set $N(x, U) := \{n \in \mathbb{N} : T^n x \in U\}$ is non-empty. We will denote by $HC(T)$ the set of hypercyclic vectors for T .

Disjointness in hypercyclicity is introduced independently by Bernal [4] and by Ràs and Paris